



Emotion-Aware Chatbot with Cultural Adaptation for Mitigating Work-Related Stress

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ABSTRACT

The accessibility to affordable and yet effective mental health support is limited due to various barriers. Given the proliferation of technology, chatbots for mental health support has been widely used. Being mindful of the users' cultural background and the ability to respond with empathy are perceived as important factors that contribute to the usability and effective communication with chatbots. Nonetheless, cultural adaptation and emotional sensitivity in mental health chatbots are not thoroughly investigated. Hence, this work aims to design and implement an emotion-aware chatbot which incorporates cultural-adaptation that could provide effective Cognitive Behavioural Therapy (CBT) interventions to Malaysian community. The emotion detection model was developed using BERT and achieved an accuracy of 0.89. For cultural adaptation, besides localised contents, Google Cloud Translation API was used as the machine translation model between Malay to English. A user study was then carried out to assess the effectiveness of emotion sensitivity and cultural adaptation in CBT-based mental health support. The ablation study shows that CBT, cultural adaptation and emotional sensitivity have positive impact on the effectiveness and usability of mental health chatbots.

CCS CONCEPTS

• **Human-centered computing** → **Interactive systems and tools**;

KEYWORDS

Mental health intervention, chatbot, emotion detection, cultural adaptation

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1 INTRODUCTION

Mental health issues have become a growing public health concern and one of the contributing factors towards global and societal burden [3]. Unfortunately, there are several barriers exist that limit the accessibility to psychological help. These barriers include but are not limited to lack of human resources to provide mental health services [7], stigma [10], low accessibility due to locations, and inability to afford treatments [11]. Hence, technology has been actively explored as a feasible solution, particularly digital conversational agents, which are commonly known as chatbots.

A systematic review on chatbots in mental health was conducted by Vaidyam et al. [23]. The preliminary evidence speaks favorably for the use of chatbots in psychiatric settings, but its effectiveness is yet to be examined thoroughly. Several works discussed the influence of chatbot empathic ability and emotional sensitivity towards user experience, as presented by Fadhil and Schiavo [9]. This indicates that emotion detection can be utilized to formulate better responses with empathy, which leads to more effective mental health chatbots. Moreover, the work by Griner and Smith [11] expressed the need for incorporating cultural adaptation, which enable more engaging user experience. Griner and Smith [11] further elaborated that appropriate types of cultural adaptations include incorporating cultural contents or values into intervention, and providing services in the native language of the user. Nonetheless, there is a lack of research assessing how these aspects can affect the effectiveness and usability of mental health chatbots. While the recently released ChatGPT performs well in online conversations, it does not provide Cognitive Behavioural Therapy (CBT) intervention.

This research project aims to investigate the effectiveness of an emotion-aware chatbot build for mental health intervention, with cultural adaptation, particularly for users in Malaysia. A text-based chatbot will be designed and implemented by incorporating conversation flow that incorporates an emotion detection model, a machine translation API, and cultural adaptation. The emotion detection model is developed using Bidirectional Encoder Representations from Transformers (BERT). The proposed chatbot aims to provide self-help intervention to mitigate work-related stress for Malaysian users. Note that the chatbot will not act to replace a psychiatrist, but rather to provide early intervention. Therefore, the effectiveness of the proposed chatbot in this research refers to how effective the self-help intervention can be delivered, instead of its effectiveness in overcoming mental health issues. The self-help intervention provided by the chatbot include various CBT-based

activities, which have been proven effective [4]. The results of an ablation study show that cultural adaptation and emotional sensitivity do impact the effectiveness of mental health chatbots.

This paper is organised as follow: Section 2 presents the related work while Section 3 details the proposed solution. Next, Section 4 presents the results and discussion and lastly, the paper is concluded in Section 5.

2 LITERATURE REVIEW

2.1 CBT Mental Health Chatbot

Several studies examined the feasibility and the effectiveness of delivering CBT interventions using text-based chatbots [10, 17, 19]. An automated conversational agent (Woebot) was used in Fitzpatrick et al. [10] to reduce self-identified depression and stress symptoms among college students. A significant reduction in depression was shown in the Woebot group compared to the information control group. Moreover, the secondary completer analysis¹ suggested that completers have a significant reduction in symptoms of anxiety. Meanwhile, an automated chatbot (Shim) was used to deliver CBT interventions for promoting well-being [17]. The complete analysis revealed substantial interaction effects of group and time on the Flourishing Scale (FS), previously known as the Psychological Well-Being Scale and the Perceived Stress Scale (PSS-10). Another study [19] used a mobile application-based interactive CBT chatbot to relieve panic disorder symptoms among patients diagnosed with panic disorder. A reduction in the severity of panic symptoms achieved by the use of the chatbot was shown more effective compared to information distribution through a paperback book. The promising results achieved in these studies validate the potential of a CBT-based mental health chatbot in facilitating mental health intervention, which motivated our research reported in this paper.

2.2 Emotional Sensitivity and Cultural Adaptation in Mental Health Chatbots

Several studies investigated the importance of emotional sensitivity or empathy in mental health chatbots. An empathetic chatbot (Wysa) was used in Inkster et al. [12] which reported a high proportion of positive in-app experiences. The chatbot (Wysa) shows empathy by responding to the emotions expressed by users throughout the conversation. Empathy is proven as an important aspect of mental health chatbot as it increases engagement and improves the user experiences. Fadhil and Schiavo [9] further added that emotional insensitivity can be counterproductive and cause users to refuse to express themselves. Another aspect that was examined related to mental health chatbot technology is cultural adaptation. Cameron et al. [5] pointed out the importance of being mindful of users' culture. The discussion about the best practices for designing mental healthcare chatbots was based on a case study on iHelp chatbot². Besides, Griner and Smith [11] added that culturally adapting mental health interventions by conducting interventions in clients' preferred language is beneficial. In addition, incorporating the cultural values of clients into therapy leads to improvement in psychotherapy utilization, retention, and outcomes. As there

¹Analyses conducted for only those patients who completed the medication/treatment phase.

²<https://core.ac.uk/download/pdf/304668034.pdf>

is a lack of research on these aspects, this research aims to examine how emotional sensitivity and cultural adaptation influence the effectiveness of mental health chatbots in providing self-help interventions.

2.3 Emotion Detection

Lexicon-based approaches, rule-based and keyword-based approaches are the earliest main approaches for emotion detection [27]. However, lexicon-based approaches are time-consuming, domain or language specific, and contain tedious processes [26]. Therefore, researchers shifted their focus to machine learning (ML) based approaches. ML-based approach requires feature engineering and feature extraction, which can be domain specific as well [26]. Hence, deep-learning approaches were examined due to their capability to create the required features for classification automatically during training, relieving the burden of feature design. A new state-of-the-art (SOTA) model was introduced in Devlin et al. [8], namely BERT. BERT achieves impressive results in many Natural Language Processing tasks [8]. Moreover, it outperforms some commonly used deep learning approaches such as long short-term memory (LSTM), convolutional neural network (CNN), recurrent neural network (RNN), gated recurrent unit (GRU), and bi-LSTM in terms of emotion detection [28]. The main strength of BERT is its capability to handle various tasks without substantial task-specific architecture modifications [8]. The promising results achieved by BERT have become a game-changer in the field of emotion detection. As BERT is the latest state-of-the-art and shows promising results, it is adopted for the implementation of emotion detection model that will be incorporated in our proposed chatbot. Section 4 discusses the performance of the models trained using different datasets in detail.

2.4 Machine Translation

The rule-based machine translation (RBMT) is one of the earliest approaches for machine translation (MT) [1]. It is reliable as it can analyze both syntax and semantic levels well to get accurate output [1]. However, it requires a great amount of linguistic knowledge and a comprehensive list of rules to cover a language completely which is impossible [6]. Therefore, researchers had shifted their attention to corpus-based approaches including statistical machine translation (SMT) and example-based machine translation (EBMT). SMT requires enormous amounts of corpus data [1] causing it to be costly for situations where limited resources are available [22]. As for EBMT, a corpus with sufficient examples containing various sentence patterns is required [1] as it only produces appropriate translations if the sentence matches an example included in the corpus [6]. Therefore, neural MT (NMT) has been introduced because it shows remarkable improvements over RBMT and SMT systems [14]. In this work, a few NMT machine translation models have been investigated. The best performing model will be incorporated in the proposed chatbot.

Some commonly used models for the NMT approach are deep neural network (DNN) and RNN. In 2017, a new NMT approach named transformer that utilized an attention mechanism was introduced by Vaswani et al. [24] and claimed as the new state-of-the-art.

The attention mechanism can cope with long input sequences effectively [25] which is the key factor in many sequence transduction tasks [24]. Vaswani et al. [24] further stated that the transformer model can train significantly faster than architectures using recurrent or convolutional layers. Moreover, it outperforms all previously reported ensembles for the Workshop on Statistical Machine Translation (WMT) 2014 English-to-German translation tasks [24]. The great results achieved have proven the potential of the transformer approach in the machine translation field.

Hybrid approaches that obtain the best possible translation by exploiting features of various existing approaches [22] are always in consideration.

3 PROPOSED SOLUTION

3.1 System Overview

Figure 1 shows the overview of the system architecture. A client application has been created to handle the integration of different modules for building the complete system. The application will start a session with the chatbot built in IBM Watson Assistant (see Section 3.4.2) when the user starts interacting with the chatbot. The emotion detection and machine translation models are contained in the Client Actions module. When the chatbot needs to perform emotion detection or text translation, it will first pause the dialog. It will then send a request for client action and the user input to the client application. Upon receiving the client action request, the application will pass the received user input to the corresponding module based on the client action name. The emotion detection or machine translation model will then process and return the detected emotion and/or translated text back to the application. Subsequently, the application will pass the client action result back to the chatbot for evaluation, and continue the dialog. The client application also handles the deployment of the chatbot to the Slack application, as discussed in Section 3.4.2. At this stage, the first prototype of this mental health chatbot is accessible through the Slack application interface. All the interactions between the users and the chatbot will need to go through the Slack interface and the client application.

3.2 Emotion Detection Model

In this work, the emotion detection model is obtained by fine-tuning a pre-trained BERT model using task-relevant datasets. Therefore, suitable benchmark datasets which reflect daily conversations were sourced. Five suitable datasets were acquired and the details are summarized in Table 1.

Various combinations of the datasets were tested in order to determine which dataset should be included in the training. Each combination generates a model and the performance will be compared. The performance comparison will be based on the accuracy achieved and the micro accuracy for each of the emotion classes. Micro accuracy is taken into consideration as the number of sentences for each emotion class in the dataset might not be balanced which might make accuracy a poor evaluation metric. Table 1 shows that each dataset is annotated with a different sets of emotions. For the purpose of this project, we included the basic six emotions that are commonly found in humans, namely happiness, sadness, anger, disgust, surprise, and fear [13]. In addition, neutral emotion is included

as well since the user responses are not necessarily subjective sentences [2, 20]. Including neutral emotion in our model is important as assigning emotion to factual statements, which are not subjective is inappropriate and will lead to unexpected behavior of the chatbot. For data pre-processing, a Python program is created to transform each dataset into a consistent format. Sentences that are not labeled with one of the seven chosen emotions will be removed. All sentences were also pre-processed to remove the unwanted characters, followed by case folding. The transformed and combined dataset was then split into training and testing datasets using 80:20 ratio. To ensure that the ratio of each emotion class is balanced in both training and testing datasets, stratified splitting is performed on each class individually.

Fine-tuning of BERT is performed on a computer with the processor, Intel Xeon W-2145 (8 cores with hyperthreading), 64GB of RAM, NVidia Quadro P5000, 512GB SSD, and 2TB HDD. The pre-processed dataset was used to fine-tune a pre-trained model in order to generate the emotion detection model. In this work, a BERT tokenizer provided by PyTorch-Pretrained-Bert library³ for “bert-base-uncased” model is used to tokenize the sentences. The pre-trained BERT classification model used is the “bert-base-uncased” model provided by the Huggingface library⁴. This model is selected as it requires lesser resources for the fine-tuning process compared to other models provided and it is not case sensitive. As for the fine-tuning parameters, different configurations can be selected as the optimum parameters vary for different datasets. In this work, the recommended settings used are stated in Table 2.

3.3 Machine Translation Model

In this work, a machine translation model using the NMT approach which can perform Malay to English translation will be used. There are several existing machine translation models available which are suitable. The machine translation model developed by Low et al. [16] fulfilled all these criteria. The technique used is the new arising transformer model and a great result is reported which is a BLEU score of 35.42. Another model is provided by IBM which is known as Watson Language Translator. It also implements the NMT approach and supports translation from Malay to English translation. However, not much information can be found regarding the technical details of the module. Besides, Google Cloud has provided a basic Translation API that uses Google’s neural machine translation technology. It supports text translation of more than 100 languages including Malay to English translation.

A comparison will be done to examine which model can achieve better translation. The comparison will be based on the BLEU score achieved by each model on the same testing data to make it fair. The testing dataset used for the BLEU evaluation is the subset of the testing dataset used in Low et al. [16]. The reason for doing so is that using Watson Language Translator and Google Cloud Translation API requires charges that are calculated based on how many characters are translated. Hence, it is unaffordable to use the full testing dataset in Low et al. [16] as the dataset is big. Therefore, only 6000 sentences are extracted randomly from it will be used as our testing data. The BLEU evaluation module to be used is the

³<https://pypi.org/project/pytorch-pretrained-bert/>

⁴https://huggingface.co/transformers/pretrained_models.html

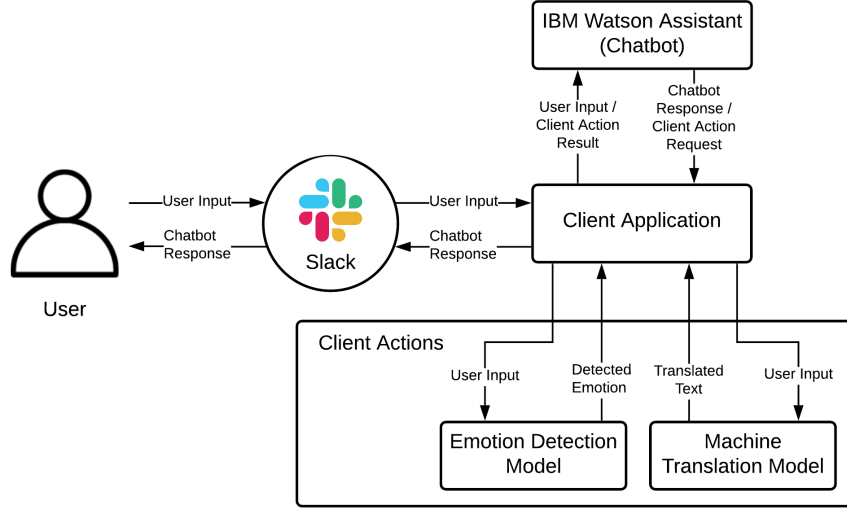


Figure 1: System architecture of the proposed emotion-aware chatbot.

No.	Datasets	Sources	Emotion Included	Dataset Size
1	SemEval-2018 Task 1: Affect in Tweets	[18]	Anger, anticipation, disgust, fear, joy, love, optimism, pessimism, sadness, surprise, trust	10983
2	Emotions dataset for NLP	[21]	Sadness, joy, fear, anger, love, surprise	20000
3	Figure Eight Labelled Textual Dataset	[2]	Anger, boredom, empty, enthusiasm, fun, happiness, hate, love, neutral, relief, sadness, surprise, worry	40000
4	DailyDialogue	[15]	no emotion, anger, disgust, fear, happiness, sadness, surprise	102968
5	MELD	[20]	Anger, disgust, fear, joy, neutral, sadness, surprise	13708

Table 1: Datasets details.

Maximum length	Batch size	AdamW-learning rate	AdamW-epsilon	Epochs
128	32	2e-5	1e-8	4

Table 2: The parameters values used for fine-tuning.

“multi-bleu-detok.perl” provided by OpenNMT⁵. Another BLEU evaluation module provided is known as “multi-bleu.perl”. However, “multi-bleu-detok.perl” is used as the scores produced do not depend on the tokenizer. The model which performed the best will be selected and integrated into the chatbot.

3.4 Emotion-Aware Chatbot

3.4.1 Conversation Flow. The conversation flow of the chatbot is first designed to determine how the chatbot will interact with the users. For the purpose of the ablation study, a total of four versions have been created for the conversation flow, which will be discussed in Section 3.5:

- Version 1: Without emotion detection and cultural adaptation
- Version 2: Without emotion detection but includes cultural adaptation
- Version 3: With emotion detection but without cultural adaptation
- Version 4: With emotion detection and cultural adaptation

The conversation flow has incorporated PHQ-9, the Patient Health Questionnaire-9 to perform a simple screening on the users’ overall depression severity. As the chatbot is incapable of providing professional treatment, the outcome of PHQ-9 enables the chatbot to converse with only users without major depression and suicidal thoughts. In the conversation flow, a total of three activities are provided, namely cognitive restructuring, relaxation and stress reduction techniques, and talk about feelings. The first two activities

⁵<https://github.com/OpenNMT/OpenNMT-py>

are CBT interventions. Cognitive restructuring involves identifying the cognitive distortions in a negative thought and reframing the thought to be more positive. As for the relaxation techniques, two exercises are included which are progressive muscle relaxation and deep breathing exercises. The third activity is to allow users to talk about their feelings and the chatbot will eventually encourage them to try out some pleasurable activities. The aim of this activity is to uplift the users' mood.

Validation of contents. A certified psychiatrist at Monash University Malaysia was engaged to validate the CBT interventions included in the chatbot, as well as the application of PHQ-9. The interventions have to be implemented correctly as the targeted users are vulnerable. The psychiatrist validated that the contents are appropriate to avoid any possibility of causing harm to the intended users.

Cultural adaptation. While designing the conversation flow, cultural adaptation was included. Some life experiences and phrases which are relatable to the majority of Malaysians are included. Moreover, another approach for cultural adaptation is to allow conversation in the users' native language. Therefore, the national language of Malaysia, i.e. Malay language was also implemented in two versions of the chatbot with cultural adaptation. Three translation models were explored for our implementation, which are neural translation model for English-Malay Low et al. [16], Watson Language Translator, and Google Cloud Translation API. From our experiments, Google Cloud Translation API outperforms the other two modules with a BLEU score of 53.96. Hence, it is chosen as the translation model.

Emotional sensitivity. Two versions of the conversation flow with emotional sensitivity are designed to utilize the emotion detection model. The emotion detection model is used to detect the emotion behind the user response when they can input freely. Based on the detected emotion, the response returned by the chatbot will be drafted to respond back appropriately. However, for situations where users have to choose one of the response options provided, the emotion detection model is not used.

3.4.2 Chatbot Development and Deployment. The conversation flow designed will be implemented as a chatbot in IBM Watson Assistant (IWA). The reasons for selecting this platform are due to the comprehensive graphical user interface provided and its security measures for data privacy. IWA also provides fine control over the dialogue processing, which supports the integration of the emotion detection and machine translation models. This can be seamlessly done by adding a client action to the dialog node where emotion detection or translation is required.

The auto-deployment provided by IWA can be used if the chatbot is deployed directly on Slack. However, as our chatbot needs to integrate with the emotion detection and translation models, a client application is needed in between the chatbot and Slack (see Section 3.1). Hence, the chatbot is deployed on Slack as a python flask application. The python flask application is the same as the client application. In this application, the slackclient library⁶ is used for interfacing with Slack Web API and Real Time Messaging

(RTM) API. The python flask application module is referred to an open-source project⁷ which did a similar deployment. Necessary modifications were done to fit the application. Figure 2 shows the screenshot of the implementation of Version 3 on Slack.

3.5 Ablation Study

An ablation study is carried out to evaluate how each component contributes to the effectiveness and usability of the chatbot. Participants are invited to participate in a user study, where the test results were analysed for the ablation study. Since the purpose of this chatbot is to alleviate work-related stress for Malaysians, a total of five Malaysians who are working adults were engaged⁸. Each participant is required to test all four versions of the chatbot. To avoid bias, the sequence of the versions to be tested are randomized for each participant, and they were not informed of the differences in each version. Before the user study was carried out, a briefing and question answering session was held to explain the task. Upon the completion of user study, the participants filled in four questionnaires, one for each version of the chatbot, and an additional overall feedback questionnaire. The questionnaire contains ten questions, as listed in Table 3. The feedback received was analysed and presented in Section 4.

4 RESULTS AND DISCUSSIONS

4.1 Performance of Emotion Detection Models

The performances of the emotion detection models tested are shown in Table 4, with the best performance in each emotion (micro accuracy), and overall accuracy highlighted in bold. Micro accuracy and overall accuracy are commonly used in emotion detection [18, 21]. The performance differences between including and excluding Tweets-like sentences in the training datasets (column 2 of Table 4) are first examined. Models M1 to M4 include datasets 1 and 3 which contain Tweets-like sentences while models M5 to M8 excluded these datasets. Models M5 to M8 achieved higher accuracy, ranging from 0.82 to 0.95, in comparison to models M1 to M4. For individual models, pairwise comparisons are done for models M1-M5, M2-M6, M3-M7, and M4-M8. These pairwise comparisons are done as the only difference between each pair is whether they included or excluded Tweets-like sentences. For all the pairwise comparisons, the models which were not trained using Tweets-like sentences have higher overall accuracy, higher micro accuracy for joy, sadness, fear, surprise, anger, and neutral. There are two exceptions, where M2 has slightly higher micro accuracy for neutral compared to M6. Another exception is M4 achieved higher micro accuracy for joy compared to M8. Overall, models which were not trained using Tweets-like sentences achieved significantly lower micro accuracy for disgust emotion in the pairwise comparison. Note that M5 does not include disgust emotion since it was trained on dataset 2 (see Table 1).

Based on the comparisons, models that were not trained on Tweets-like sentences achieve better overall accuracy despite weak performance in detecting the disgust emotion. The reason is because Tweets-like sentences are normally incomplete or contain

⁶<https://pypi.org/project/slackclient/>

⁷<https://github.com/IBM/watson-online-store>

⁸Approval from Monash University Human Research Ethics Committee was obtained before we started the user study (Project ID: 30480)

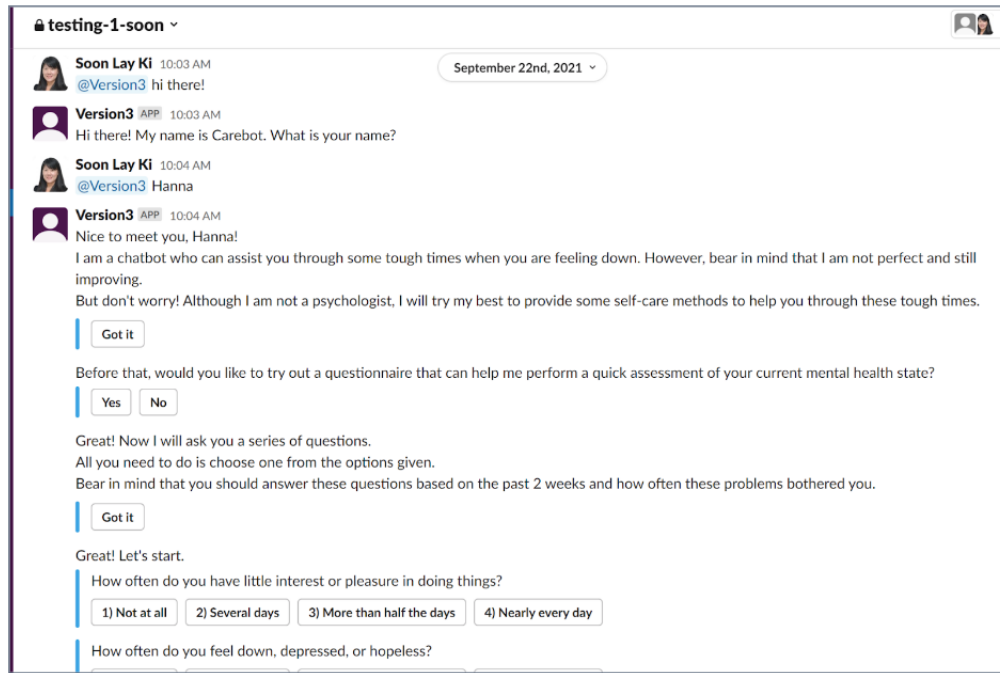


Figure 2: Implementation of Chatbot on Slack (Version 3)

No	Question	Type
1	Do you think the chatbot is aware of your emotion?	Yes or No
2	If yes, what makes you think so?	Free text
3	Do you think the chatbot includes Malaysia cultural adaptation?	Yes or No
4	If yes, what makes you think so?	Free text
5	How enjoyable are the activities?	Scale of 1 to 5 or Not applicable
6	How do you feel after completing each activity?	Scale of 1 to 3 or Not applicable
7	Which aspect of the chatbot do you like?	Multiple choice
8	Which aspect of the chatbot do you like?	Multiple choice
9	On a scale of 1 to 5, how helpful do you think this chatbot is?	Scale of 1 to 5
10	Do you have any suggestions to improve this chatbot?	Free text

Table 3: Questions in the user study questionnaire.

hashtags, user tagging, or abbreviations. Models which were not trained on Tweets-like sentences, namely M6, M7 and M8 were further investigated. Neutral emotion is included since it is commonly found in conversations, as discussed in Section 3.2. Since the micro accuracy for disgust emotion is low for models that were not trained using Tweets-like sentences, disgust is excluded from the emotion labels for our the second round of model training. Likewise, the micro accuracy for surprise emotion is also relatively low compared to other emotions. Although surprise emotion is one of the basic emotions, it can be excluded from the emotion labels because it is less common for mental health intervention. To support our hypothesis, we trained M6 to M8 again but excluding disgust and surprise emotions. The new models are M9 to M11, with their performance shown in the second part of Table 4. M9 to M11 achieved better performance compared to models M6 to M8.

Therefore, disgust and surprise emotions can be excluded for better accuracy. The final comparison is between model M9 to M11. Based on the results, M9 achieved the highest accuracy, micro accuracy for fear, anger, and neutral emotion among all. Meanwhile, M10 obtained the highest micro accuracy for joy and sadness emotion. As M9 has better micro accuracy for the majority of the emotions and highest overall accuracy, it is chosen as the emotion detection model for our chatbot.

BERT is suitable for emotion detection tasks. By performing fine-tuning on a pre-trained BERT model, an emotion detection model with high accuracy is obtained with a training time of fewer than two hours. This shows that BERT is suitable for emotion detection tasks since a good result is obtained in a short time. Hence, even a larger dataset is used for the fine-tuning process in the future, it is feasible to do so by using GPU. Moreover, there are no substantial

No.	Datasets	Micro Accuracy							Accuracy
		Joy	Sadness	Disgust	Fear	Surprise	Anger	Neutral	
M1	1, 2, 3	0.8829	0.8344	0.7052	0.8438	0.3619	0.8428	-	0.80
M2	1, 2, 3, 4	0.7388	0.7606	0.6194	0.8349	0.4530	0.7333	0.9350	0.85
M3	1, 2, 3, 5	0.8583	0.7931	0.6210	0.7806	0.4502	0.6759	0.7925	0.76
M4	1, 2, 3, 4, 5	0.7313	0.7406	0.5713	0.7151	0.4683	0.6121	0.9198	0.82
M5	2	0.9837	0.9620	-	0.9137	0.7639	0.9114	-	0.95
M6	2, 4	0.7414	0.8409	0.1408	0.8585	0.6654	0.7735	0.9294	0.88
M7	2, 5	0.8826	0.8912	0.0278	0.8168	0.6964	0.7451	0.7941	0.82
M8	2, 4, 5	0.7076	0.7862	0.1678	0.7539	0.5861	0.6404	0.9214	0.85
M9	2, 4	0.7406	0.8567	-	0.9116	-	0.7721	0.9332	0.89
M10	2, 5	0.8853	0.8897	-	0.8626	-	0.7555	0.8260	0.85
M11	2, 4, 5	0.7231	0.7912	-	0.8227	-	0.6592	0.9257	0.87

Table 4: Performance of the emotion detection models.

No.	Datasets	Micro Accuracy					Accuracy
		Joy	Sadness	Fear	Anger	Neutral	
M9	2, 4	0.7406	0.8567	0.9116	0.7721	0.9332	0.89
M10	2, 5	0.8853	0.8897	0.8626	0.7555	0.8260	0.85
M11	2, 4, 5	0.7231	0.7912	0.8227	0.6592	0.9257	0.87

Table 5: Performance of the additional tested emotion detection models.

task-specific architecture modifications needed which make it easy to be implemented. Furthermore, even with no hyperparameters fine-tuning, it had performed well. This shows that BERT has a huge potential in achieving higher accuracy for emotion detection if the hyperparameters are set appropriately and a better dataset is used.

4.2 Performance of Machine Translation Model

No.	Models	BLEU score
1	Model by Low et al. [16]	40.00
2	Watson Language Translator	28.33
3	Google Cloud Translation API	53.96

Table 6: BLEU score achieved by each machine translation model.

As shown in Table 6, the model which performed the best is Google Cloud Translation API which achieved a BLEU score of 53.96. However, we are unable to investigate what causes such results for these models as we do not have access to the training data used for each model. Without knowing the nature of the training dataset, it is hard to analyze what factors have affected the performance. Hence, we try comparing the performance of these models with several examples. For the sentence "Saya berasa penat kerana terlalu banyak kerja.", the translations given by each model are shown in the following:

- Model 1: I am tired of being too kerja.
- Model 2: I feel tired of being too much work.
- Model 3: I feel tired from too much work.

Another sentence tested is "Saya cuma nak relax sikit and enjoy" and the results are shown in the following:

- Model 1: I just want to take a little relax and enjoy
- Model 2: I just want to relax a little and enjoy
- Model 3: I just want to relax a little and enjoy

Based on the two examples, the translation provided by model 3 is the most appropriate among all. For the first example, model 1 did not perform well as it did not even translate the word "kerja" correctly and the meaning of the translated sentence is incorrect. The sentence translated by model 2 is also incorrect. As for the second example, the translated sentence is correct and the same as the output by model 3. However, the sentence translated by model 1 is incorrect. Hence, based on the BLEU score and the examples, Google Cloud Translation API will be used as the machine translation model to be integrated into the chatbot.

However, the actual potential of these models on handling texts used in conversations between users and mental health chatbots is unable to be examined. This is due to the absence of such a dataset in this domain. If such a dataset exists in the future, it can be used to improve the suitability of these models for our application and also improve their performance. Hence, further examination is required to explore in detail how these models perform in our application.

NMT has achieved great results. All three models included in the comparison are NMT models. The BLEU scores achieved by these model ranges from 28.33 to 53.96 which has a big difference. Hence, this indicates that even the same approach is used, the performance might be different. The reasons are that the implementation might be different and the datasets used have a huge impact on the performance. However, these results had shown the potential of NMT in handling Malay to English translation.

4.3 Ablation Study

CBT-based Contents. Based on the feedback collected through the user study questionnaire, the enjoyability level reported for all three CTB-based activities is similar and consistent across all four versions of the chatbot. The participants also stated that they either felt the same or better after completing each activity. This indicates that the contents of the chatbot are helpful in mitigating work-related stress and do not bring negative effects to the users. As these contents include three CBT interventions, this result shows that CBT is suitable for mental health chatbots which can provide self-help guidance, without the supervision of professional psychiatrists. The full potential of CBT in mental health chatbots can be further explored where more suitable activities can be included in the future version.

Influence of Emotional Sensitivity. The participants were not aware of whether the version they are testing are emotionally aware or not. If their answer for Question 1 is yes, a follow-up question that asks why do they think so is included. Surprisingly, for versions that are not emotion sensitive, the majority of the participants mistaken these versions are emotionally aware. The reasons are that the personal experiences shared by the chatbot relate to the users' actual experiences and the chatbot shared stories describing its own emotion. These caused the participants to mistaken personification as emotion awareness. Hence, the effect of emotional sensitivity with regard to the effectiveness and usability of the chatbot should be further examined. Nonetheless, one of the feedback stated that version 4 chatbot can read responses better. This shows that the version with emotion detection is able to formulate a better response. On the contrary, one feedback for version 3 has stated that the chatbot did not sense the correct emotion in the user response. The response given by the participant is "better" for Question 6. However, the chatbot responds with "It is alright to feel the same...". Upon further investigation, the chatbot responded as such due to the detected neutral emotion. By right, the chatbot should have detected the joy emotion instead. This shows that the emotion detection model used contains some flaws despite having achieved high accuracy. This also explains why only a few participants found the versions with emotion detection give better responses since the emotions detected might be incorrect. This feedback provides some evidence that providing emotionally sensitive responses can impact the usability and effectiveness of the chatbot. Nonetheless, the impact is much reliant on the accuracy of the emotion detection model. Based on the experimental results, our emotion detection model will be devised in the future to further improve its effectiveness in the application of mental health chatbots.

Influence of Cultural Adaptation. For versions that did not include cultural adaptation, one of the suggestions given by participants is to include a Malay version. These suggestion might be given due to the participants tested version with cultural adaptation earlier. However, this situation shows that chatbot that includes cultural adaptation is preferred by the users. One of the feedback given for version 3 is that the participant felt like getting information from the chatbot only without much engagement. This is mainly due to

a lack of examples relating to Malaysian culture. Another participant suggested to include Chinese in the future. One of the good aspects of the version 2 chatbot, as pointed out by a participant is that it has different language versions. These feedback show the importance of cultural adaptation in mental health chatbots, particularly in enabling conversation in their native language. For future improvement, we plan to engage more native Malay speakers to assist in the proofreading the Malay contents. Similar to emotional sensitivity, the participants were not informed of which version includes cultural adaptation during the user study. For the two versions with cultural adaptation, four out of five participants had indicated that they think these versions are culturally adapted. The reasons are Malay version is provided, some common terms such as "*kopitiam*" and experiences like "attending SPM" in their culture are used⁹. This indicates that cultural adaptation is not limited to only implementing chatbots in the users' native language but also including things related to their culture and life experiences. Overall, both the quantitative and qualitative feedback provided with regard to the effectiveness of cultural adaptation is positive. In summary, the analysis on the feedback collected through the user study shows that emotion detection is important in providing effective responses with empathy while cultural adaptation enables the users to relate to the CBT activities better.

Potential Issue with Emotion Detection Model Used. The potential issues with the emotion detection model used are examined. An interesting observation is found that is the DailyDialogue dataset (dataset 4) used for fine-tuning this model is heavily imbalanced. The number of sentences for each emotion in dataset 4 is shown in Table 7.

As shown in Table 7, the majority of the sentences belong to neutral emotion. This explains why the micro accuracy for neutral emotion is relatively higher which is 0.9332 as the model might be biased towards the neutral emotion. To verify this, another model is developed by using the same set of datasets as the model we used but with downsampling performed on the neutral class for dataset 4 and tested. The performance of this newly created model is shown in Table 8.

By comparing both models, the newly created model has better micro accuracy for joy, sadness, fear, and anger emotion. However, it achieved a lower micro accuracy for neutral emotion and the overall accuracy. This result shows that the new model has a more balanced performance which can reduce bias towards the neutral emotion despite having lower accuracy. Therefore, the potential cause for the bad performance of the emotion detection model used is it might be biased towards the neutral emotion. Hence, in the future, it is better to use a more balanced model.

Chatbot with more flexibility is favorable. A question asking whether users prefer to input freely or choose from options is included in the overall feedback questionnaire. Four out of five participants had selected that they would prefer to input freely over choosing a response from the options provided by the chatbot. This shows a strong preference for input freely flexibility while chatting with the chatbot.

⁹SPM is one of the common examinations in Malaysia.

Emotion	Joy	Sadness	Disgust	Fear	Surprise	Anger	Neutral
No.	12885	1150	353	174	1823	1022	85572
Total	102979						

Table 7: Number of sentences for each emotion in dataset 4 (DailyDialogue).

Datasets included	Micro Accuracy					Accuracy
	Joy	Sadness	Fear	Anger	Neutral	
2, 4 (down sampled)	0.8310	0.8826	0.9155	0.7962	0.8625	0.85

Table 8: Performance of the newly created emotion detection models.

Moreover, some feedbacks have stated that the sensitivity to capture the user response should be improved. Most of the time, the designed chatbot will provide options for the user to choose as their response. If only one option is provided and the user input something else, the chatbot will assume that the user had selected the only option provided. Therefore, this behavior causes the users to feel ignored if they reply to the chatbot with a question when one option is provided. Additionally, some feedbacks stated that the bad aspect is the predefined responses which makes them feel like it is not a real chat. This shows that always providing options to the users might not be ideal and the user experience is affected negatively.

Suggestions for future improvement. One of the suggestion given is to include some pictorial explanations to deliver the information better. In this era, there are several elements that are commonly used in text conversations such as emojis, gifs, images, and videos. By including these elements in the chatbot, the interactions will be more interesting which can enhance the user experience of using the chatbot. Moreover, enabling the use of emojis can be beneficial as the emotion behind the user response can be evaluated based on the emojis. This can further improve the chatbot to be more emotionally aware to form better responses. Another suggestion is to include gratitude journalling options to change negative thoughts. One of the CBT interventions is journalling to gather a person's moods and thought. Hence, this intervention can be examined and included in the chatbot. As there are feedbacks stating that the content of the chatbot is repetitive, including more CBT interventions and extending the current conversation flow will be beneficial. Therefore, a mental health chatbot should contain more functionalities if possible and suitable.

5 CONCLUSION

This paper presents our proposed emotion-aware chatbot with cultural adaptation for mental health intervention. The incorporated emotion detection model (M9) performs well with an accuracy of 0.89. As part of the cultural adaptation, we implemented Google Cloud Translation API as it achieved the best performance with a BLEU score of 53.96. A comprehensive ablation study was conducted based on the quantitative and qualitative feedback collected through a user study. Four different versions of the chatbots were assessed in the user study. The analyses show that cultural adaptation can improve user engagement and user experience, which enables more effective mental health interventions. The chatbot is

currently undergoing a migration to mobile application platform. It is anticipated that the implementation on a mobile device can provide more realistic experience.

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